



Robot Programming Final Project

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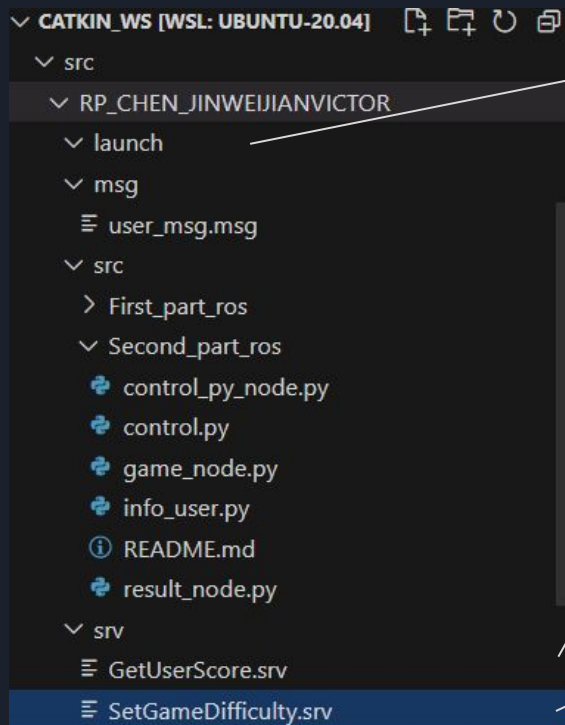


Introduction

- Implementation of a classic snake game using ROS nodes.
- Inter node communication via:
 - Subscribers
 - Services



General Folder disposition



RP_CHEN_JINWEIJIANVICTOR

launch

game_control.launch

game_pygame.launch

src > RP_CHEN_JINWEIJIANVICTOR > srv > GetUserScore.srv

```
1 string username
2 ---
3 int64 score
```

src > RP_CHEN_JINWEIJIANVICTOR > srv > SetGameDifficulty.srv

```
1 string change_difficulty
2 ---
3 bool success
```

Game node

```
class GameNode:
    def __init__(self):
        """Initialize the game node with ROS and Pygame"""
        # Initialize ROS node
        rospy.init_node('game_node')

        # Initialize ROS parameters with defaults
        self.user_name = rospy.get_param('~user_name', 'default_user')
        self.player_color = rospy.get_param('~change_player_color', 1) # 1:Red, 2:Purple, 3:Blue
        self.screen_param = rospy.get_param('~screen_param', 'phase1')

        # ROS Subscribers
        self.user_sub = rospy.Subscriber('user_information', user_msg, self.user_callback)
        self.control_sub = rospy.Subscriber('keyboard_control', String, self.control_callback)

        # ROS Publishers
        self.score_pub = rospy.Publisher('result_information', Int64, queue_size=10)
        self.phase_pub = rospy.Publisher('game_phase', String, queue_size=10)

        # ROS Services
        self.score_service = rospy.Service('user_score', GetUserScore, self.handle_get_score)
        self.difficulty_service = rospy.Service('difficulty', SetGameDifficulty, self.handle_set_difficulty)
```

- Parameters created

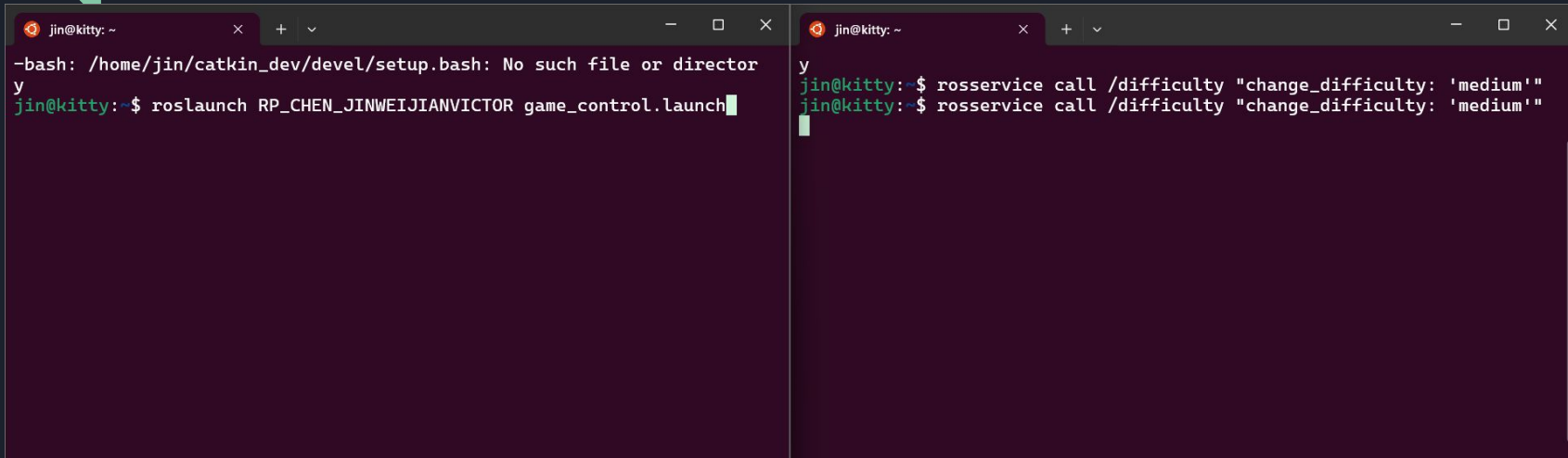
- Subscribers
- Publishers
- Services

Launcher (control without pygame)

```
src > RP_CHEN_JINWEIJIANVICTOR > launch > game_control.launch
1 <!--game_control.launch (Using regular control node) -->
2 <launch>
3   <!-- Game node with parameters (no separate terminal needed) -->
4   <node name="game_node" pkg="RP_CHEN_JINWEIJIANVICTOR" type="game_node.py" output="screen">
5     <param name="user_name" value="default_user" type="string"/>
6     <param name="change_player_color" value="1" type="int"/>
7     <param name="screen_param" value="phase1" type="string"/>
8   </node>
9
10  <!-- Info user node (separate terminal for input) -->
11  <node name="info_user" pkg="RP_CHEN_JINWEIJIANVICTOR" type="info_user.py" output="screen">
12    launch-prefix="xterm -hold -title 'User Input' -e"/>
13
14  <!-- Control node (separate terminal for control) -->
15  <node name="control_node" pkg="RP_CHEN_JINWEIJIANVICTOR" type="control.py" output="screen">
16    launch-prefix="xterm -hold -title 'Game Control' -e"/>
17
18  <!-- Result node (separate terminal for results) -->
19  <node name="result_node" pkg="RP_CHEN_JINWEIJIANVICTOR" type="result_node.py" output="screen">
20    launch-prefix="xterm -hold -title 'Game Results' -e"/>
21 </launch>
```

- Parameters need to be defined in launcher.
- Selected text means that its opening a new terminal for the selected node when the launcher is run.

Launcher functionality

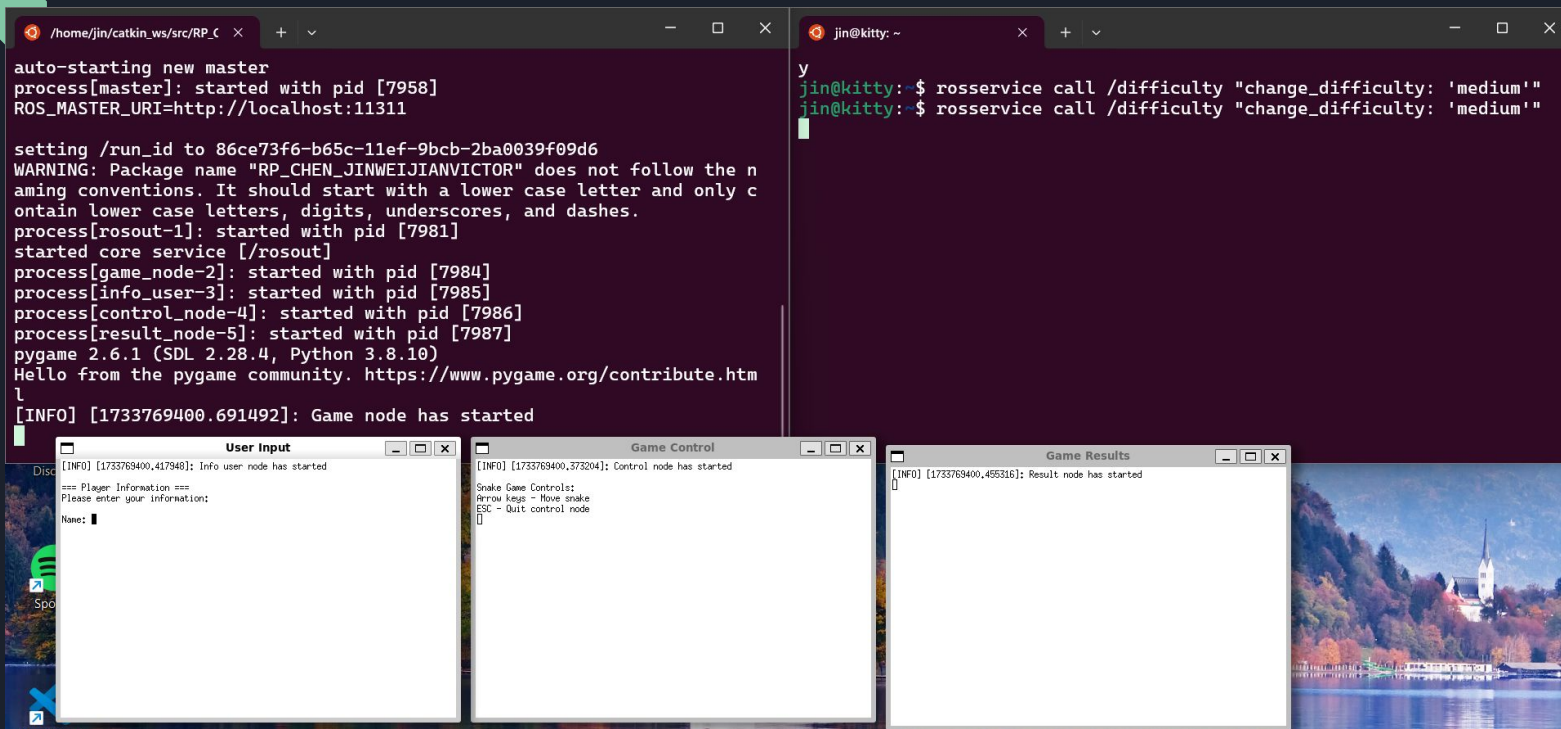


The image shows two terminal windows side-by-side. The left window shows a user attempting to run a ROS launch file, receiving an error about a missing setup.bash file, and then successfully launching a game control package. The right window shows the user calling a ROS service to change the difficulty level to 'medium'.

```
jin@kitty: ~  
-bash: /home/jin/catkin_dev/devel/setup.bash: No such file or director  
y  
jin@kitty:~$ roslaunch RP_CHEN_JINWEIJIANVICTOR game_control.launch  
  
jin@kitty:~$ rosservice call /difficulty "change_difficulty: 'medium'"  
jin@kitty:~$ rosservice call /difficulty "change_difficulty: 'medium'"
```

- No roscore needed
- Bashrc edited so it sources automatically
- Before launching and calling service

Launcher started

The image shows a desktop environment with a dark blue background and a light blue and green geometric design on the left. In the center, there are three terminal windows. The top-left window is titled '/home/jin/catkin_ws/src/RP_C' and shows the output of a ROS launch script. The top-right window is titled 'jin@kitty: ~' and shows the execution of rosservice calls. The bottom row contains three smaller windows: 'User Input', 'Game Control', and 'Game Results'. The 'User Input' window prompts for player information. The 'Game Control' window lists snake game controls. The 'Game Results' window shows the result node has started. A scenic image of a lake and a church is visible in the bottom right corner.

```
/home/jin/catkin_ws/src/RP_C x + v
auto-starting new master
process[master]: started with pid [7958]
ROSMaster_URI=http://localhost:11311

setting /run_id to 86ce73f6-b65c-11ef-9bcb-2ba0039f09d6
WARNING: Package name "RP_CHEN_JINWEIJIANVICTOR" does not follow the naming conventions. It should start with a lower case letter and only contain lower case letters, digits, underscores, and dashes.
process[rosout-1]: started with pid [7981]
started core service [/rosout]
process[game_node-2]: started with pid [7984]
process[info_user-3]: started with pid [7985]
process[control_node-4]: started with pid [7986]
process[result_node-5]: started with pid [7987]
pygame 2.6.1 (SDL 2.28.4, Python 3.8.10)
Hello from the pygame community. https://www.pygame.org/contribute.html
[INFO] [1733769400.691492]: Game node has started

jin@kitty: ~ x + v
y
jin@kitty:~$ rosservice call /difficulty "change_difficulty: 'medium'"
jin@kitty:~$ rosservice call /difficulty "change_difficulty: 'medium'"

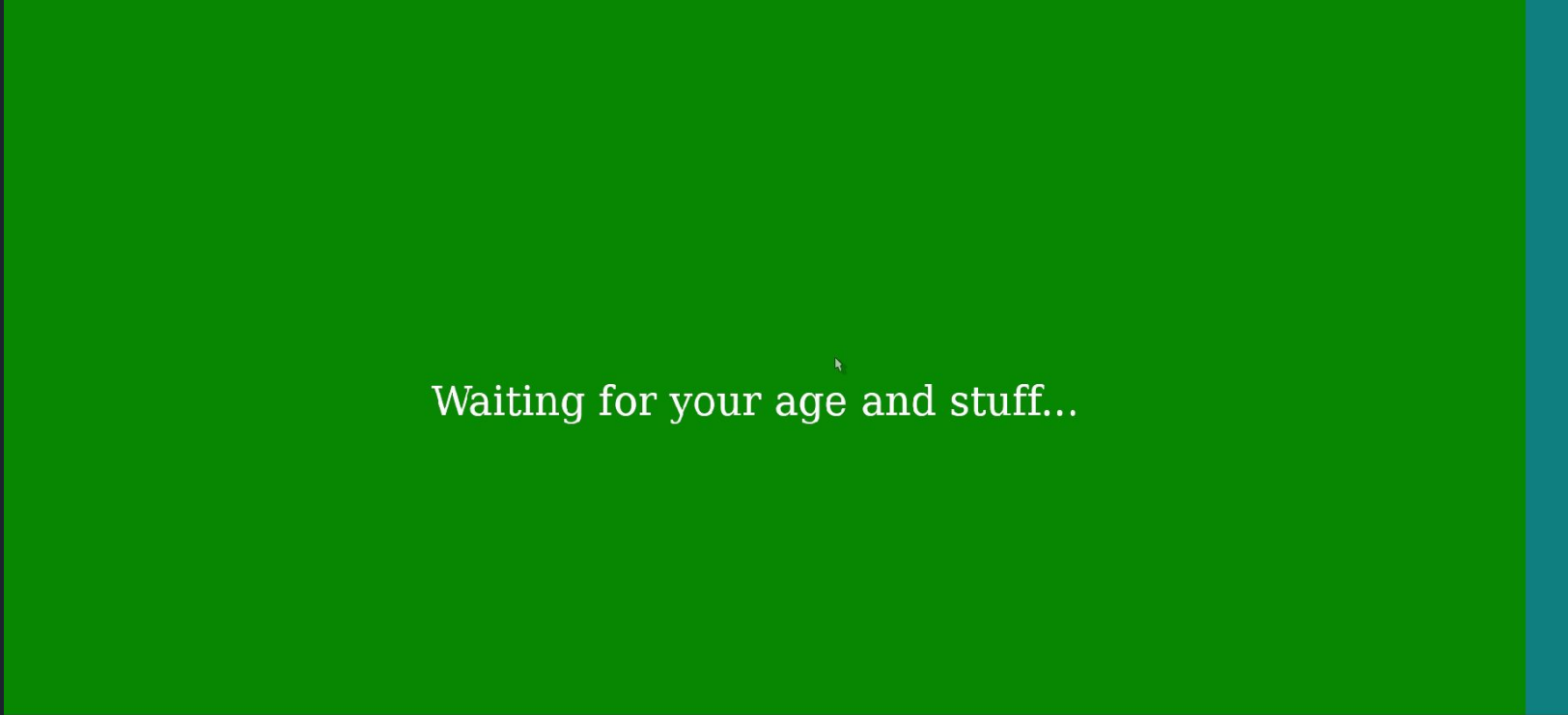
[INFO] [1733769400.417948]: Info user node has started
=== Player Information ===
Please enter your information:
Name: 

[INFO] [1733769400.373204]: Control node has started
Snake Game Controls:
Arrow keys - Move snake
ESC - Quit control node

[INFO] [1733769400.455316]: Result node has started
```



Game window



Waiting for your age and stuff...

Subscriptions working

```
User Input  
[INFO] [1733769400.417948]: Info user node has started  
=== Player Information ===  
Please enter your information:  
Name: victor  
Username: kitty  
Age: 21
```

Welcome kitty of age 21 ;) <3)!

Select your snakes favorite color!!

1 - Red

2 - Purple

3 - Blue

Welcome kitty of age 21 ;) <3)!

Select your snakes favorite color!!

- 1 - Red
- 2 - Purple
- 3 - Blue

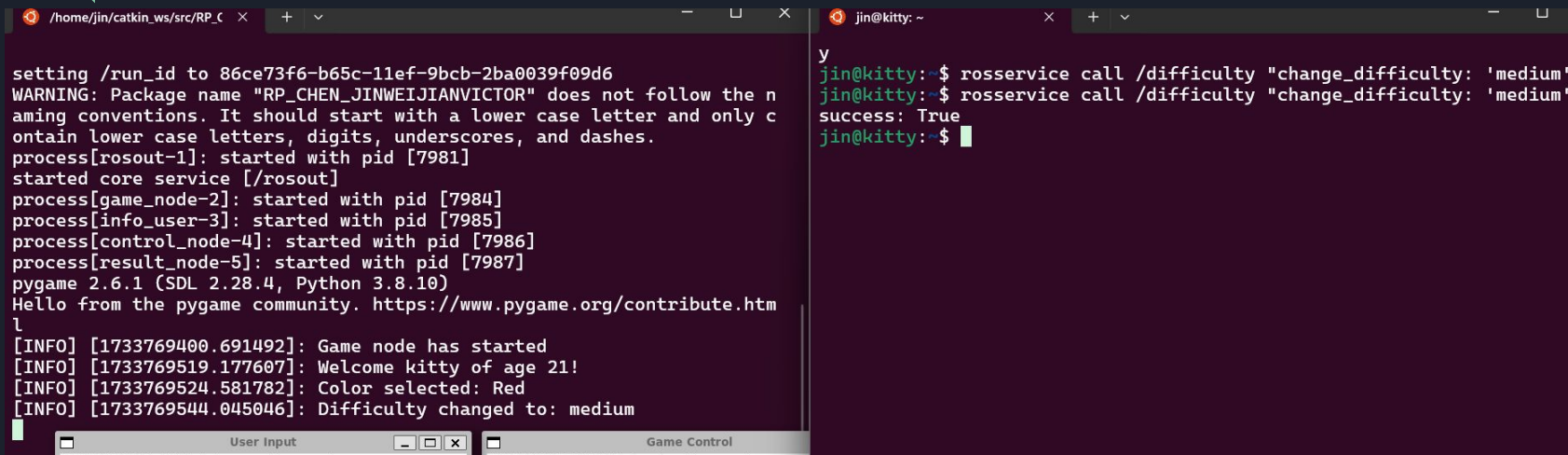
After selecting color...



Welcome kitty of age 21 ;) <3)!

Waiting for difficulty selection...
(Please use service call to set difficulty)

Service Requested

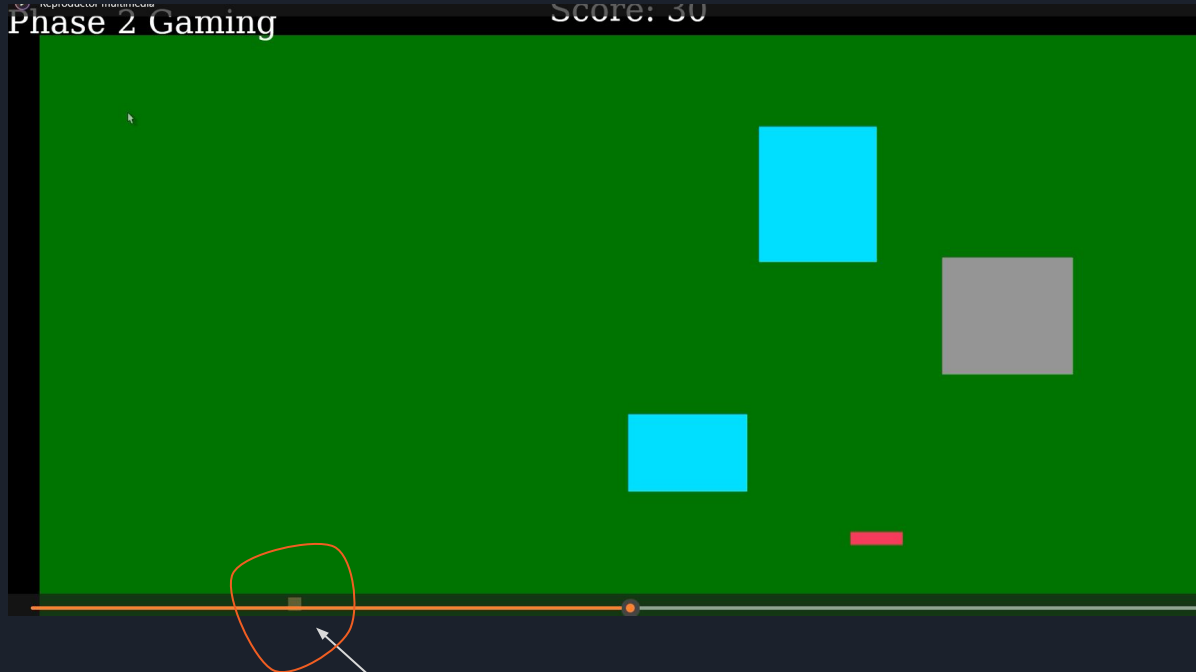


The image shows two terminal windows. The left window, titled '/home/jin/catkin_ws/src/RP_C', displays the output of a ROS node startup. It shows a warning about package naming conventions, followed by the successful startup of several processes (rosout, game_node-2, info_user-3, control_node-4, result_node-5) with their respective PIDs. It also shows the pygame version (2.6.1) and a welcome message for 'kitty' of age 21. The difficulty is set to 'medium'. The right window, titled 'jin@kitty: ~', shows two rosservice calls to '/difficulty' with the parameter 'change_difficulty: 'medium'', both of which were successful. Below the terminals, there are two small windows: 'User Input' and 'Game Control'.

```
setting /run_id to 86ce73f6-b65c-11ef-9bcb-2ba0039f09d6
WARNING: Package name "RP_CHEN_JINWEIJIANVICTOR" does not follow the n
aming conventions. It should start with a lower case letter and only c
ontain lower case letters, digits, underscores, and dashes.
process[rosout-1]: started with pid [7981]
started core service [/rosout]
process[game_node-2]: started with pid [7984]
process[info_user-3]: started with pid [7985]
process[control_node-4]: started with pid [7986]
process[result_node-5]: started with pid [7987]
pygame 2.6.1 (SDL 2.28.4, Python 3.8.10)
Hello from the pygame community. https://www.pygame.org/contribute.htm
l
[INFO] [1733769400.691492]: Game node has started
[INFO] [1733769519.177607]: Welcome kitty of age 21!
[INFO] [1733769524.581782]: Color selected: Red
[INFO] [1733769544.045046]: Difficulty changed to: medium

y
jin@kitty:~$ rosservice call /difficulty "change_difficulty: 'medium'
jin@kitty:~$ rosservice call /difficulty "change_difficulty: 'medium'
success: True
jin@kitty:~$
```

- In the ubuntu terminal you can see anything that is happening !!
- Colors selected and difficulty



- The grey areas corresponds to where the next obstacle is going to spawn.
- They appear each time a treat is eaten (yellow)



Game over panel

Game Over! Final Score: 30

Press R to Restart, E to Exit, H for homepage

- 3 options in case you wanna continue gaming
- Restart doesn't change Snake color or difficulty

```
de.py", line 15, in <module>
    exec(compile(fh.read(), python_script, 'exec'), context)
File "/home/jin/catkin_ws/src/RP_CHEN_JINWEIJIANVICTOR/src/Second_pa
rt_ros/game_node.py", line 509, in <module>
    game_node.run()
File "/home/jin/catkin_ws/src/RP_CHEN_JINWEIJIANVICTOR/src/Second_pa
rt_ros/game_node.py", line 476, in run
    if self.handle_keyboard_events():
File "/home/jin/catkin_ws/src/RP_CHEN_JINWEIJIANVICTOR/src/Second_pa
rt_ros/game_node.py", line 391, in handle_keyboard_events
    self.cleanup_and_exit()
AttributeError: 'GameNode' object has no attribute 'cleanup_and_exit'
[game_node-2] process has died [pid 7984, exit code 1, cmd /home/jin/c
atkin_ws/devel/lib/RP_CHEN_JINWEIJIANVICTOR/game_node.py __name:=game_
node __log:=/home/jin/.ros/log/86ce73f6-b65c-11ef-9bcb-2ba0039f09d6/ga
me_node-2.log].
log file: /home/jin/.ros/log/86ce73f6-b65c-11ef-9bcb-2ba0039f09d6/game
_node-2*.log
```

```
jin@kitty: ~
y
jin@kitty:~$ rosservice call /difficulty "chang
jin@kitty:~$ rosservice call /difficulty "chang
success: True
jin@kitty:~$
```

```
User Input
[INFO] [1733769400,417948]: Info user node has started

=== Player Information ===
Please enter your information:

Name: victor
Username: kitty
Age: 21
[INFO] [1733769619,175935]: User information sent: victor, kitty, 21
```

```
Game Control
[INFO] [1733769400,373204]: Control node has started

Snake Game Controls:
Arrow keys - Move snake
ESC - Quit control node
```

```
Game Results
High Score: 30
Games Played: 17
Average Score: 1.76

[INFO] [1733769576,655806]: Final Score for kitty: 30
[INFO] [1733769576,734201]:
Score Analysis for kitty:
[INFO] [1733769576,736487]: Score Percentage: 30%

Score Report:
Player: victor (kitty)
Score Percentage: 30%

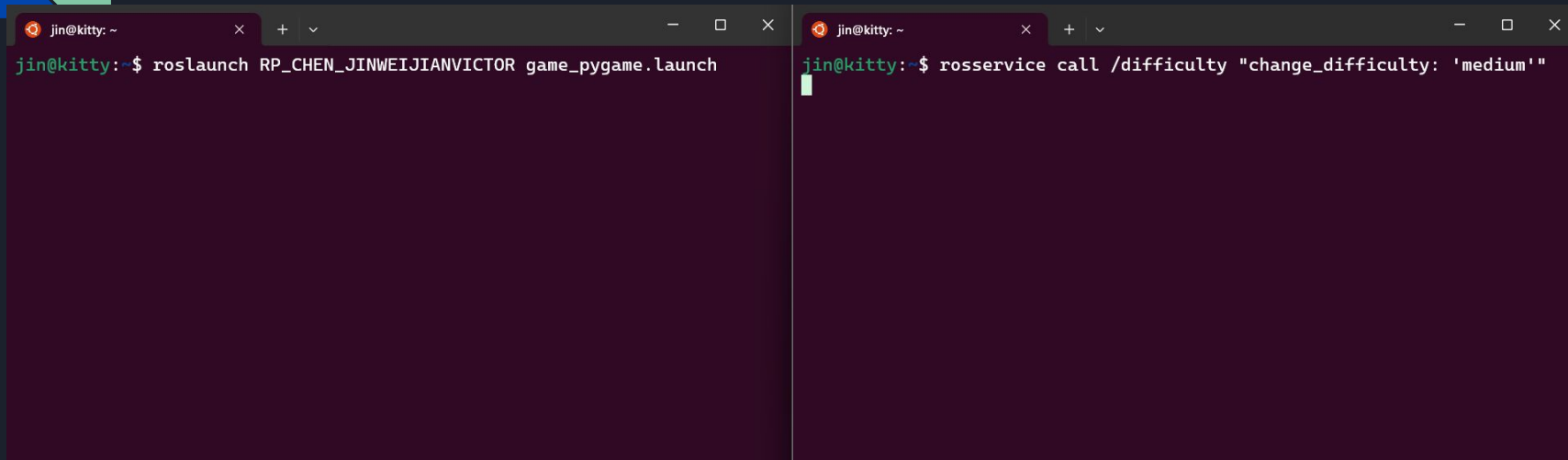
Game Over! Final Report:

Player: victor (kitty)
Final Score: 30
Performance: 30%
High Score: 30
Games Played: 18
Average Score: 1.67

[INFO] [1733769576,739034]: Final Score for kitty: 30
```

- You can see the game results even when you exit the game because other nodes are still running

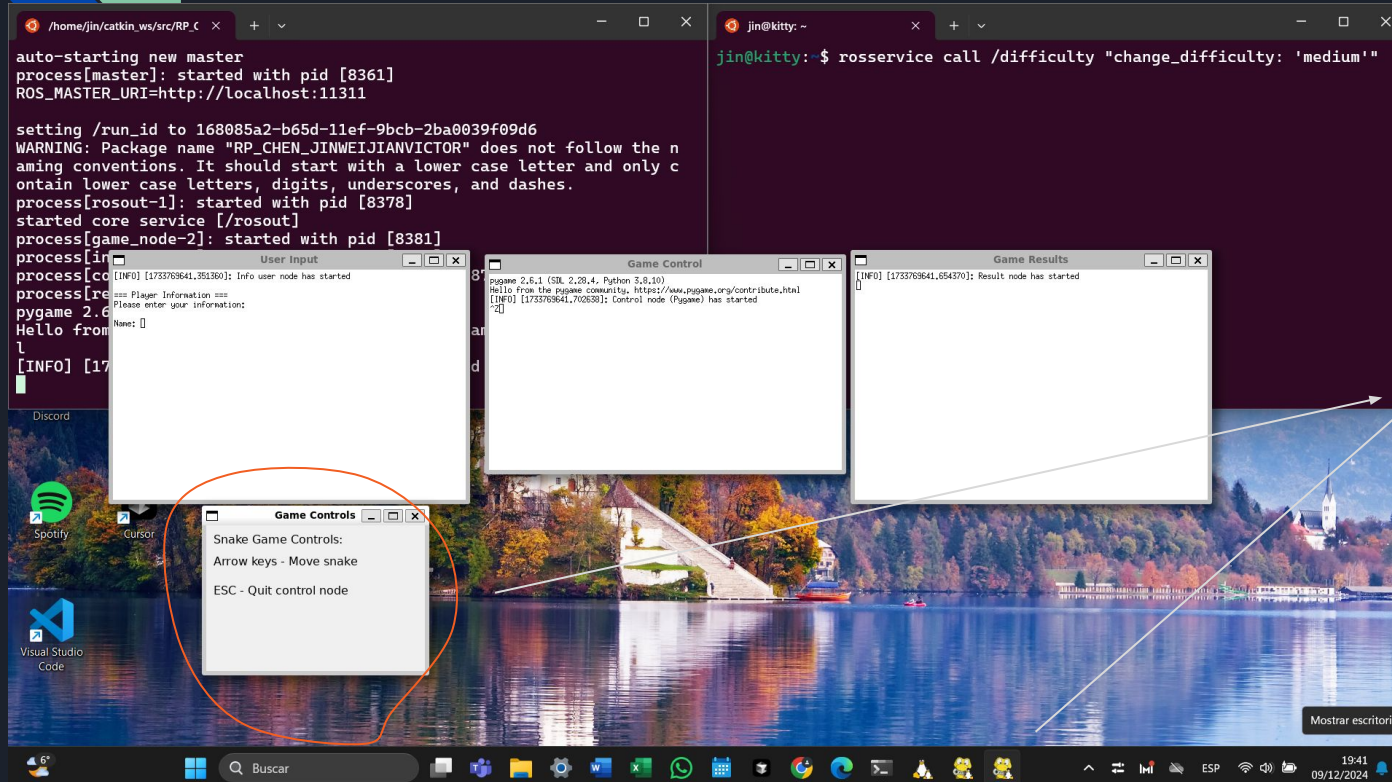
Launcher with pygame



```
jin@kitty: ~  
jin@kitty:~$ roslaunch RP_CHEN_JINWEIJIANVICTOR game_pygame.launch  
  
jin@kitty:~$ rosservice call /difficulty "change_difficulty: 'medium'"
```

Essentially the same launcher as with control node except that the node called in launcher is the other one

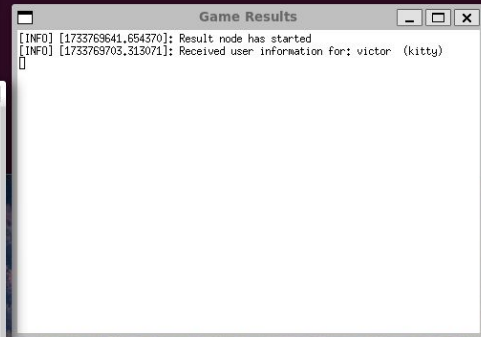
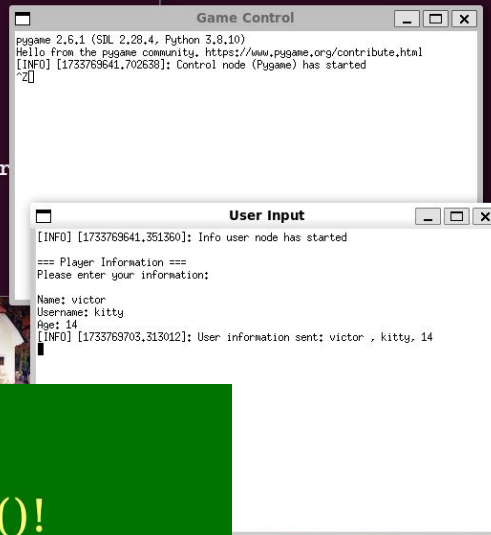
Key difference



- There are now two pygame windows

```
roprocess[rosout-1]: started with pid [8378]
tarted core service [/rosout]
roprocess[game_node-2]: started with pid [8381]
roprocess[info_user-3]: started with pid [8382]
roprocess[control_py_node-4]: started with pid [8387]
roprocess[result_node-5]: started with pid [8388]
pygame 2.6.1 (SDL 2.28.4, Python 3.8.10)
ello from the pygame community. https://www.pygame.org/contribute.html
```

```
INFO] [1733769641.674829]: Game node has started
INFO] [1733769703.313063]: Welcome kitty of age 14!
```



Welcome kitty of age 14 :():!

Select your snakes favorite color!!

- 1 - Red
- 2 - Purple
- 3 - Blue

- Game results receives user information
- Sad face if your underage
- Same procedure as before

Game paused panel



- Multiple options plus you can see the background and know where the snake is when you unpause (press P)



Thank You!



Together, we're changing how people view and treat snakes